

Rosemary

SCIENTIFIC NAME

Rosmarinus officinalis

FAMILY

Lamiaceae/Labiatae

OTHER COMMON NAMES

Alecrim, Compass Plant, Compass Weed, Encensier, Herbe Aux Couronnes, Old Man, Polar Plant, Romarin, Romarin Des Troubadours, Romero, Rose de Marie, Rose Des Marins, Rosée De Mer, Rosemarine, Rusmari, Rusmary

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Overview

Rosemary is a perennial, evergreen shrub native to the Mediterranean region and grown worldwide ([71821](#), [91730](#)). Traditionally, rosemary has been used orally for gastric disorders, as well as for cough, anxiety, depression, gout, headache, insomnia, and liver

and gallbladder complaints. It has also been used topically for pain, skin conditions like eczema, and as an insect repellent.

Safety

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LIKELY SAFE when used orally in amounts typically found in foods. Rosemary has Generally Recognized as Safe (GRAS) status in the US ([4912](#)).

POSSIBLY SAFE when the leaf is used orally and appropriately in medicinal amounts ([18](#)). Powdered rosemary leaf has been used with apparent safety as a single dose of up to 1.5 grams ([18246](#), [91731](#)) or at a dose of 1-4 grams daily for up to 8 weeks ([91727](#), [98536](#), [105327](#), [109561](#)). ...when the essential oil is used topically and appropriately for up to 7 months ([5177](#), [91729](#), [109560](#)). ...when the essential oil is used by inhalation as aromatherapy, short-term ([7107](#), [18323](#), [105324](#), [109559](#)).

LIKELY UNSAFE when the essential oil or very large quantities of rosemary leaf are used orally. Ingestion of undiluted rosemary oil or very large quantities of rosemary leaf can cause serious adverse effects ([18](#), [515](#)).

PREGNANCY: POSSIBLY UNSAFE when used orally in medicinal amounts. Rosemary might have uterine and menstrual flow stimulant effects ([4](#), [12](#), [18](#)), and might increase metabolism of estradiol and estrone ([18331](#)); avoid using. There is insufficient reliable information available about the safety of rosemary when used topically during pregnancy.

LACTATION: There is insufficient reliable information available about the safety of using rosemary in medicinal amounts during lactation; avoid using.

Adverse Effects

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General

Orally, rosemary seems to be well tolerated when used in appropriate medicinal amounts. Undiluted rosemary oil or very large quantities of rosemary leaf should not be consumed. Topically and as aromatherapy, rosemary seems to be well tolerated.

Dermatologic

Topically, rosemary use can lead to photosensitivity, erythema, dermatitis, and cheilitis in hypersensitive individuals ([4](#), [6](#)).

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Immunologic

Topically, allergic reactions can occur. When used in the mouth, lip and gum edema have occurred ([101173](#)). When used on the skin, allergic contact dermatitis has occurred, likely due to the constituent carnosol ([71715](#), [71924](#), [71926](#)).

Rosemary might also cause occupational asthma. A case of occupational asthma caused by several aromatic herbs including thyme, rosemary, bay leaf, and garlic has been reported. The diagnosis was confirmed by inhalation challenges. Although all of the herbs caused immediate skin reactivity, a radioallergosorbent test (RAST) showed that garlic was the most potent allergen by weight, with rosemary and the other herbs showing less reactivity ([783](#)).

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Neurologic/CNS

Orally, the undiluted oil, as well as the camphor constituent of rosemary, might cause seizures ([4](#), [5](#), [6](#), [12868](#)).

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Effectiveness

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POSSIBLY EFFECTIVE

Memory. Oral rosemary seems to improve memory in young adults. It is unclear if rosemary aromatherapy improves memory.

Clinical research in healthy adults shows that taking rosemary 500 mg orally twice daily for one month improves memory by approximately 14% when compared with placebo ([98536](#)). Other preliminary clinical research shows that applying four drops

of pure rosemary essential oil (Tisserand Aromatherapy) to an aromatherapy diffuser pad 5 minutes before a single test period can improve the quality, but not the speed, of memory recall, and increase alertness and contentment more than lavender aromatherapy or no aroma ([18323](#)).

Rosemary aromatherapy has also been evaluated for improving memory in adults with kidney failure. A small clinical study in patients (mean age 53 years) undergoing hemodialysis shows that aromatherapy with rosemary essential oil three times weekly for 1 month does not improve medication adherence or measures of prospective and retrospective memory when compared with a control group. In this study, five drops of rosemary essential oil were applied to gauze, which was then placed 10 cm from the patient's nose for 30 minutes starting 1 hour after hemodialysis ([109559](#)).

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INSUFFICIENT RELIABLE EVIDENCE TO RATE

Abortion. Although there has been interest in the use of oral rosemary as an abortifacient, there is insufficient reliable information about the clinical effects of rosemary for this purpose.

Age-related cognitive decline. It is unclear if oral rosemary is beneficial for age-related cognitive decline.

A small clinical study in healthy individuals aged 65-90 years shows that a single 750 mg dose of powdered rosemary leaf in tomato juice may improve memory speed. However, higher single doses of 1500-6000 mg seem to have a deleterious effect on memory speed. For other measures of attention and memory, there were no clear benefits across the range of doses from 750-6000 mg ([18246](#)).

Oral rosemary has also been evaluated in combination with other ingredients. A small clinical study in healthy adults aged 62 years or younger shows that taking a combination product containing equal parts rosemary, lemon balm, and sage twice

daily for 2 weeks modestly improves word recall when compared with placebo. However, it does not appear to have benefit in adults aged 63 years and older ([97277](#)). It is unclear if any benefit is due to rosemary, other ingredients, or the combination.

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Alopecia areata. Topical rosemary oil has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

Preliminary clinical research shows that rosemary oil 114 mg, in combination with lavender oil 108 mg, thyme oil 88 mg, and cedarwood oil 94 mg, mixed with jojoba oil 3 mL and grapeseed oil 20 mL and applied topically to the scalp, improves hair growth in 44% of patients, compared with 15% of those receiving placebo. This combination was massaged into the scalp for a minimum of 2 minutes, followed by wrapping the head in a warm towel, every night for 7 months ([5177](#)).

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Androgenic alopecia. It is unclear if topical rosemary is beneficial for androgenic alopecia.

Preliminary clinical research shows that applying rosemary oil (Barij Essence Pharmaceutical Company) 1 mL to the scalp twice daily for 6 months is as effective as minoxidil 2% for increasing hair count in patients with androgenic alopecia ([91729](#)).

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Depression. It is unclear if oral rosemary is beneficial for major depressive disorder.

A small clinical trial among adults newly diagnosed with major depressive disorder and recent initiation of antidepressant medication shows that taking rosemary dried leaf extract 650 mg twice daily for 8 weeks might reduce some symptoms of depression and anxiety when compared with placebo ([112141](#)). However, the validity of these findings is limited by differences in baseline depression symptom scores between groups.

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Diabetes. It is unclear if oral rosemary is beneficial for improving glycemic control in patients with type 2 diabetes.

In patients with type 2 diabetes, preliminary clinical research shows that taking rosemary tea daily for 90 days decreases glycated hemoglobin levels by 15% and waist and hip circumference by 6% and improves insulin resistance. However, there was no effect on fasting blood glucose, lipid profile, body weight, or body mass index (BMI). The tea was made by adding rosemary 2 grams to a liter of boiling water for 3 minutes and then passing through a sieve ([105323](#)). Other preliminary clinical research in adults with type 2 diabetes shows that taking rosemary powder 3 grams daily for 4 weeks does not reduce fasting blood glucose levels or improve lipid profile when compared with baseline. These endpoints were improved in a group of healthy adults ([105327](#)). The validity of the findings from these studies is limited by the lack a comparator group.

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Diabetic nephropathy. Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

Preliminary clinical research shows that taking a specific combination product (Canephron N, Bionorica) containing 18 mg each of rosemary leaf, centaury herb, and lovage root per tablet, as two tablets three times daily for 6 months along with standard antidiabetes treatment and enalapril (Vasotec), decreases microalbuminuria by 73%, decreases albumin:creatinine ratio by 46%, increases high-density lipoprotein (HDL) cholesterol by 25%, and lowers triglyceride levels by 43%, but does not improve glomerular filtration rate, when compared to baseline. Improvements were greater in those taking the combination product than in those treated only with standard antidiabetes therapy and enalapril. However, the combination product does not appear to improve reductions in total cholesterol or low-density lipoprotein (LDL) cholesterol ([91726](#)).

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Dysmenorrhea. It is unclear if oral rosemary is beneficial for dysmenorrhea.

In patients with moderate primary dysmenorrhea, clinical research shows that taking rosemary extract 220 mg every 8 hours for the first three days of the menstrual cycle, repeated for two menstrual cycles, is as effective as mefenamic acid for reducing average pain and bleeding when compared with two baseline cycles ([105328](#)).

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Fatigue. Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

Preliminary clinical research in healthy young adults with below-average energy levels shows that taking a single dose of a mixture of rosemary from Albania (*Rosmarinus officinalis*) and Morocco (*Rosmarinus eriocalyx*) 1.7 grams does not consistently improve sustained attention, motivation to perform cognitive tasks, mental energy, or mental fatigue when compared with placebo ([91731](#)).

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Fibromyalgia. Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

A small clinical study in patients with fibromyalgia shows that taking an oral product containing rosemary leaf extract, oleanolic acid from olive leaf, and rho iso-alpha acids from hops (Meta050, Metagenics), at a dose of 440 mg three times daily for 4 weeks, then 880 mg twice daily for 4 weeks, does not improve symptoms when compared to baseline ([18327](#)).

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Gingivitis. A mouth rinse containing rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

Preliminary clinical research in adults with mild to moderate gingivitis and mild plaque shows that rinsing the mouth with 10 mL of mouthwash containing 5% v/v of a combined alcoholic extract of rosemary, calendula, and ginger for 30 seconds twice daily after meals for 2 weeks decreases plaque formation, gingival inflammation, and gingival bleeding similarly to chlorhexidine gluconate 0.2% mouthwash and better than a placebo mouthwash ([93555](#)).

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Hypertension. Although there is interest in using oral rosemary for hypertension, there is insufficient reliable information about the clinical effects of rosemary for this condition.

Hypotension. It is unclear if oral rosemary is beneficial for hypotension.

Preliminary clinical research in patients with primary hypotension shows that taking rosemary essential oil (Metapharmaceutical) 1 mL every 8 hours for 44 weeks increases systolic blood pressure by 13 mmHg and diastolic blood pressure by 7 mmHg when compared to baseline. However, blood pressure returned to near baseline values after treatment discontinuation ([91730](#)).

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Mental alertness. It is unclear if rosemary aromatherapy is beneficial for improving mental alertness.

Preliminary clinical research in nurses working on a night shift shows that placing one drop of rosemary oil on a surgical mask that is worn for 2 hours with a 10-minute on/15-minute off cycle modestly increases alertness and reduces sleepiness when compared with a water drop control ([105324](#)).

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Nonalcoholic fatty liver disease (NAFLD). It is unclear if oral rosemary leaf is beneficial in patients with NAFLD.

A small clinical study in patients with NAFLD shows that taking rosemary leaf powder 2 grams twice daily for 8 weeks, in conjunction with diet and exercise recommendations, does not improve measures of liver function, glycemic control, cholesterol, or body weight when compared with placebo ([109561](#)).

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Opioid withdrawal. It is unclear if oral rosemary is beneficial for opioid withdrawal when used in combination with methadone.

Preliminary clinical research shows that taking rosemary leaf along with methadone for 2 weeks improves symptoms of opioid withdrawal and the ability to sleep when compared with methadone alone after 3 and 7 days of treatment, but not after 2 weeks. Rosemary leaf was provided in capsules containing 300 mg each (Barij Essence Pharmaceutical Company) at a dose of 16 capsules daily for the first 3 days, 12 capsules daily for the next 4 days, and eight capsules daily for the next 7 days, in divided doses ([91727](#)).

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Osteoarthritis. Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

Preliminary clinical research shows subjective reductions in pain associated with osteoarthritis when oral formulations containing rosemary leaf extract, oleanolic acid from olive leaf, and rho iso-alpha acids from hops (NG440-2, Metagenics; Meta050, Metagenics) are used ([18326](#), [18327](#)). The formulation used in one of these studies (Meta050, Metagenics) was taken in a dose of 440 mg three times daily for 4 weeks, then 880 mg twice daily for 4 weeks ([18327](#)). The formulation used in the other study, which was taken daily for 4 weeks, contained rosemary leaf extract 112.5 mg, oleanolic acid from olive leaf 1 mg, and rho iso-alpha acids from hops 225 mg per tablet ([18326](#)).

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Raynaud syndrome. It is unclear if topical rosemary essential oil is beneficial in patients with Raynaud syndrome.

A very small, open-label study in patients with Raynaud syndrome caused by systemic scleroderma shows that applying 10 drops of rosemary 10% essential oil to the hands does not increase skin temperature or improve warmth perception when compared with olive oil ([109560](#)). However, this study may have been inadequately powered to detect a difference between groups.

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Rheumatoid arthritis (RA). Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear. Preliminary clinical research shows subjective decreases in pain associated with RA when oral formulations containing rosemary leaf extract, oleanolic acid from olive leaf, and rho iso-alpha acids from hops (NG440-2, Metagenics; Meta050, Metagenics) are used ([18326](#), [18327](#)). The formulation used in one of these studies (Meta050, Metagenics) was taken in a dose of 440 mg three times daily for 4 weeks, then 880 mg twice daily for 4 weeks ([18327](#)). The formulation used in the other study, which was taken daily for 4 weeks, contained rosemary leaf extract 112.5 mg, oleanolic acid from olive leaf 1 mg, and rho iso-alpha acids from hops 225 mg per tablet ([18326](#)).

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Stress. It is unclear if rosemary aromatherapy is beneficial for stress. Preliminary clinical research on the use of rosemary aromatherapy for anxiety and stress is conflicting. In a group of graduate nursing students, perceived stress associated with taking a test was lower when inhalers containing rosemary or lavender oil were used. Three drops of rosemary essential oil were added to an inhaler and inhaled before and during the test. Pulse rates, but not blood pressure, were also reduced ([18324](#)). In contrast, a small clinical study in adults aged 18-30 years shows that the aroma from diluted rosemary essential oil applied to the wrist during timed crossword puzzle completion actually increases subjective feelings of anxiety and tension when compared with placebo ([18325](#)).

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Sunburn. Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

Preliminary clinical research shows that taking a combination of rosemary and grapefruit extracts in a 1:1 ratio (NutroxBio, Montelore, Inc.), 250 mg orally once daily for 12 weeks, increases the amount of UV radiation needed to cause sunburn by 34% after 8 weeks and 56% after 12 weeks when compared to baseline ([91728](#)). The validity of these findings is limited by the lack of a comparator group.

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Upper respiratory tract infection (URTI). Oral rosemary has only been evaluated in combination with other ingredients; its effect when used alone is unclear.

A small clinical trial among healthy adults shows that taking a combination product containing rosemary leaf extract, aloe gel powder, and poria mushroom extract daily for 8 weeks does not reduce the frequency, duration, or severity of URTI-related symptoms when compared with placebo. However, in patients receiving an influenza vaccine midway through the study, this product does seem to improve some immune response biomarkers when compared with placebo ([111921](#)). The amount of rosemary leaf extract in the supplement was not disclosed.

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Wound healing. It is unclear if topical rosemary cream is beneficial for wound healing.

A moderate-sized clinical trial among primiparous adults with medio-lateral episiotomy shows that applying a cream containing rosemary extract to the wound twice daily for 10 days improves episiotomy wound healing and reduces redness and discharge when compared with placebo cream ([112143](#)).

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More evidence is needed to rate rosemary for these uses.

Dosing & Administration

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Adult

Oral:

Rosemary has been used as an oil, powder, or extract. See [Effectiveness](#) section for condition-specific information.

Taking rosemary extracts might reduce the absorption of non-heme iron from foods by 15% to 20% ([8110](#)). The clinical significance of this effect is unknown.

Topical:

Rosemary has been used as an oil, cream, or mouthwash. See [Effectiveness](#) section for condition-specific information.

Inhalation (Aromatherapy): Rosemary oil has been added to an inhaler, gauze, or diffuser pad. See [Effectiveness](#) section for condition-specific information.

Standardization & Formulation

Rosemary products are often standardized to polyphenolic content or gallic acid equivalents. In one clinical study, rosemary extract was standardized to about 57 mg of gallic acid equivalents per 100 grams ([91728](#)). In another clinical study, a mixture of rosemary from Albania (*Rosmarinus officinalis*) and Morocco (*Rosmarinus eriocalyx*) was standardized to contain rosmarinic acid 20 mg/gram. The total phenolic content was standardized to 49.8 mg of gallic acid equivalents per gram ([91731](#)).

In another clinical trial, rosemary oil (Metapharmaceutical) was standardized to contain 1,8-cineol 47.6%, camphor 13.8%, and alpha-pinene 11.7% ([91730](#)).

A topical formulation of rosemary oil (Barij Essence Pharmaceutical Company) has been standardized to contain 1,8-cineole 3.7 mg per mL ([91729](#)).

Interactions with Drugs



[EXPAND ALL](#) | [COLLAPSE ALL](#)

ANTICOAGULANT/ANTIPLATELET DRUGS

[Interaction Rating](#) MODERATE Be cautious with this combination.

[Severity](#) HIGH [Occurrence](#) POSSIBLE

Level of Evidence D (In vitro or animal study)

Theoretically, rosemary may increase the risk of bleeding if used with anticoagulant or antiplatelet drugs.

In vitro and animal research suggests that rosemary inhibits platelet aggregation ([18334](#), [18335](#), [18336](#), [18337](#)).

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ANTIDIABETES DRUGS

Interaction Rating MODERATE Be cautious with this combination.

Severity MODERATE **Occurrence** POSSIBLE

Level of Evidence B (Nonrandomized clinical trial)

Theoretically, taking rosemary with antidiabetes drugs might increase the risk of hypoglycemia.

Animal research shows that rosemary extract can decrease blood glucose levels in diabetic models ([71821](#), [71923](#)). However, research in humans is conflicting.

Although rosemary powder decreased blood glucose levels in healthy adults ([105327](#)), no change in blood glucose levels was seen in adults with type 2 diabetes, most of whom were taking antidiabetes drugs ([105323](#), [105327](#)).

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ASPIRIN

Interaction Rating MODERATE Be cautious with this combination.

Severity MODERATE **Occurrence** POSSIBLE

Level of Evidence D (In vitro or animal study)

Theoretically, rosemary might have additive effects with salicylate-containing drugs such as aspirin.

Rosemary is reported to contain salicylates ([18330](#)).

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CHOLINE MAGNESIUM TRISALICYLATE (Trilisate)

[Interaction Rating](#) MODERATE Be cautious with this combination.

[Severity](#) MODERATE **[Occurrence](#)** POSSIBLE

[Level of Evidence](#) D (In vitro or animal study)

Theoretically, rosemary might have additive effects with salicylate-containing drugs such as choline magnesium trisalicylate.

Rosemary is reported to contain salicylate ([18330](#)).

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CYTOCHROME P450 1A1 (CYP1A1) SUBSTRATES

[Interaction Rating](#) MINOR Be watchful with this combination.

[Severity](#) MODERATE **[Occurrence](#)** UNLIKELY

[Level of Evidence](#) D (In vitro or animal study)

Theoretically, rosemary might decrease the levels and clinical effects of CYP1A1 substrates.

In vitro research shows that rosemary induces CYP1A1 enzymes ([18332](#), [18333](#)).

This effect has not been reported in humans.

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CYTOCHROME P450 1A2 (CYP1A2) SUBSTRATES

[Interaction Rating](#) MINOR Be watchful with this combination.

[Severity](#) MODERATE **[Occurrence](#)** UNLIKELY

[Level of Evidence](#) D (In vitro or animal study)

Theoretically, rosemary might decrease the levels and clinical effects of CYP1A2 substrates.

In vitro research shows that rosemary induces CYP1A2 enzymes ([18332](#), [18333](#)).

This effect has not been reported in humans.

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SALSALATE (Disalcid)

[Interaction Rating](#) MODERATE Be cautious with this combination.

[Severity](#) MODERATE **[Occurrence](#)** POSSIBLE

[Level of Evidence](#) D (In vitro or animal study)

Theoretically, rosemary might have additive effects with salicylate-containing drugs such as salsalate.

Rosemary is reported to contain salicylate ([18330](#)).

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Interactions with Supplements

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[EXPAND ALL](#) | **[COLLAPSE ALL](#)**

ANTICOAGULANT/ANTIPLATELET HERBS AND SUPPLEMENTS

Theoretically, rosemary may have antiplatelet effects.

In vitro and animal research shows that rosemary inhibits platelet aggregation ([18334](#), [18335](#), [18336](#), [18337](#)). Theoretically, concomitant use of rosemary with herbs that might affect platelet aggregation could increase the risk of bleeding in some people. See other products with antiplatelet activity [here](#). See products with anticoagulant activity [here](#).

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HERBS AND SUPPLEMENTS WITH HYPOGLYCEMIC POTENTIAL

Theoretically, rosemary might have hypoglycemic effects.

Animal research shows that rosemary extract can decrease blood glucose levels in diabetic models ([71821](#), [71923](#)). However, research in humans is conflicting.

Although rosemary powder decreased blood glucose levels in healthy adults ([105327](#)), no change in glycemic control was seen in adults with type 2 diabetes, most of whom were using antidiabetes drugs ([105323](#), [105327](#)). Theoretically, rosemary might have additive effects with herbs that decrease blood glucose levels. See other products with hypoglycemic activity [here](#).

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IRON

Theoretically, rosemary might reduce the absorption of iron supplements.

Rosemary extracts reduce absorption of non-heme iron from foods by 15% to 20% ([8110](#)). This may also occur with iron supplements, although the clinical significance of this effect is unclear.

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SALICYLATE-CONTAINING HERBS

Theoretically, concomitant use of rosemary and other herbs that contain salicylate might result in additive effects.

Rosemary is reported to contain salicylates ([18330](#)).

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Interactions with Conditions

[×](#)

[EXPAND ALL](#) | [COLLAPSE ALL](#)

BLEEDING DISORDERS

Theoretically, rosemary may prolong bleeding time and increase the risk of bruising and bleeding. In vitro and animal research shows that rosemary can inhibit platelet aggregation ([18334](#), [18335](#), [18336](#), [18337](#)).

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SALICYLATE ALLERGY

Rosemary contains salicylates ([18330](#)). Individuals with salicylate allergy should use rosemary with caution.

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SEIZURE DISORDERS

Theoretically, rosemary might potentiate seizure activity due to its cineole and camphor content ([7107](#), [12868](#)). Avoid using amounts greater than those found in foods.

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Interactions with Lab Tests

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None known.

Overdose

[×](#)

Presentation

Orally, misuse of very large amounts of leaves containing rosemary oil might cause deep coma, spasm, vomiting, gastroenteritis, uterine bleeding, kidney irritation, pulmonary edema, and death ([18](#)). Ingestion of undiluted rosemary oil might cause stomach and intestinal irritation, kidney damage, and seizures ([5](#), [6](#)). Additionally, the camphor constituent of rosemary can sometimes lead to seizures ([4](#)).

Treatment

There is insufficient reliable information available about the treatment of overdose with rosemary.

Pharmacokinetics

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Absorption

In humans, the rosemary constituent luteolin, as well as derivatives of various constituents, were found in the plasma following consumption of rosemary tea

([105325](#)). Animal research shows that the absolute bioavailability of rosmarinic acid following topical administration is 60% ([71890](#)).

Distribution

Distribution of rosmarinic acid following intravenous administration in rat has been described by a two-compartment open model. One half-hour after injection, rosmarinic acid was detected and measured in brain, heart, liver, lung, muscle, spleen, and bone tissue ([71890](#)). Four and a half hours after topical administration, rosmarinic acid was measured in blood, skin, muscle, and bone tissue ([71890](#)).

Metabolism

In humans, some constituents of rosemary, including rosmarinic acid, are metabolized by colonic microbiota to metabolites such as caffeic acid and 3,4-dihydroxyphenyl-lactic acid. Caffeic acid and other constituents or metabolites are metabolized in plasma to glucuronidated or sulfated phase II derivatives ([105325](#)).

Excretion

In humans, rosemary constituents, including rosmarinic acid and caffeic acids, as well as derivatives of various constituents, were found in the urine following consumption of rosemary tea ([105325](#)). In an animal study, following the intravenous administration of rosmarinic acid, the elimination half-life was 1.8 hours ([71890](#)).

Mechanism of Action

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General

The applicable part of rosemary is the leaf. The active constituent of the leaf is the essential oil. The dried leaves contain 0.6% to 2.5% of the essential oil ([4](#), [5](#), [18246](#)). The oil consists primarily of 1,8-cineole, borneol, camphor, carvacrol, and alpha-pinene ([4](#), [6](#), [18246](#)). Other constituents include phenolic diterpenes, flavones, the caffeic acid derivative rosmarinic acid, the triterpene ursolic acid, other terpenes such as verbenone, and carnosic acid ([18246](#), [71757](#), [98532](#), [98540](#), [105325](#), [105326](#)).

Anti-inflammatory effects

In vitro research shows that rosemary extract can inhibit the activity of human leukocyte elastase, which plays a role in inflammation ([71949](#)). Other in vitro research shows that the rosemary constituent rosmanol inhibits lipopolysaccharide (LPS)-induced induction of nitric oxide synthase (iNOS) and cyclooxygenase (COX)-2 expression ([71872](#)). The rosemary constituents carnosic acid and carnosol appear to

inhibit the formation of pro-inflammatory leukotrienes in polymorphonuclear leukocytes ([71832](#)). Ursolic acid from rosemary is thought to inhibit the activation of nuclear transcription factor kappaB ([31374](#)). Some animal research suggests that intratracheal exposure to rosemary extract can prevent allergic airway inflammation by inhibiting the expression of interleukin (IL)-13 ([71765](#)). However, in human research, rosemary had no effect on plasma levels of most inflammatory mediators ([105326](#)).

Anticancer effects

Animal research suggests that rosemary extract can decrease carcinogen-induced tumor incidence, tumor burden, and tumor yield ([71782](#), [71785](#), [71914](#)). The rosemary extract and some constituents, such as carnosic acid, appear to inhibit the proliferation and possibly induce apoptosis of leukemia cells ([71699](#), [71804](#), [71836](#)). Other in vitro research suggests that rosmarinic acid, a constituent of rosemary, reduces the expression of the pro-inflammatory gene cyclooxygenase-2 (COX-2). This gene is thought to be a risk factor for cancer development ([71844](#)). Rosemary extract and rosemary essential oil may also reduce cancer development by protecting DNA from oxidative damage; this is attributed to phenolic constituents of rosemary ([71672](#), [71694](#), [71698](#)). In addition, rosemary polyphenols may reduce the activation of procarcinogens and increase detoxification pathways ([71920](#)). Other cell culture research suggests that the rosemary constituent carnosol inhibits cancer cell metastasis by decreasing the activity of metalloproteinase-9 (MMP-9) ([71751](#)).

Anticoagulant/antiplatelet effects

There is preliminary evidence that rosemary inhibits platelet aggregation ([18334](#), [18335](#), [18336](#), [18337](#)).

Antidiabetic effects

Some animal research shows that intramuscular injections of rosemary leaf oil can increase plasma glucose and decrease insulin secretion in healthy rabbits ([71910](#)). However, other animal research shows that rosemary extract can decrease blood glucose levels and increase serum insulin concentrations in diabetic rabbits and mice ([71821](#), [71923](#)). Cell culture research suggests that rosemary extract may lower blood glucose levels by activating peroxisome proliferator-activated receptor gamma (PPARgamma). The effect is attributed to the constituents carnosol and carnosic acid ([71783](#)). Other cell culture research shows that rosemary may have alpha-glucosidase inhibitory activity. This effect is attributed to the constituent rosmarinic acid ([67226](#)).

Antimicrobial effects

Various cell culture studies show that rosemary essential oil and rosemary extract have antibacterial activity against Gram negative and Gram positive bacteria, including some drug-resistant strains and food-borne pathogens, as well as activity against some species of yeast and fungus including *Candida albicans* ([13453](#), [31330](#), [43430](#), [43486](#), [71717](#), [71739](#), [71757](#), [71774](#), [71798](#), [71814](#))([71833](#), [71837](#)). Rosemary essential oil also appears to inhibit the of growth and aflatoxin production in *Aspergillus parasiticus* ([46242](#)). Rosemary essential oils with the highest quantity of camphor, borneol, and verbenone constituents appear to exert the highest antimicrobial activity ([71757](#)). Rosemary extract containing carnosic acid, carnosol, and rosmarinic acid appears to have the highest antimicrobial activity ([71774](#)). Some in vitro research suggests that rosemary extract can inhibit bacteria biofilm adhesion or biofilm production ([68505](#), [71833](#), [71840](#)). However, other research suggests that rosemary does not display antifungal activity against *Candida* species ([52026](#)). Furthermore, some research has shown that rosemary extract can act as a substrate for mycelial growth, sporulation, and aflatoxin production ([39827](#)).

Antinociceptive effects

Rosemary oil extract administration in mice reduced the number of writhing movements induced by the intraperitoneal administration of acetic acid solution and inhibited licking and shaking behaviors in both the early (neurogenic pain) and late (inflammatory pain) phases of the formalin test ([71791](#)). These effects may be modulated by the serotonin system (i.e., 5-HT1A receptors) or the endogenous opioid system ([71853](#)).

Antiobesity effects

Evidence from animal research shows that rosemary leaf extract may prevent weight and fat mass gain in mice fed a high-fat diet by inhibiting pancreatic lipase activity ([71875](#)). Cell culture research shows that the rosemary constituents carnosic acid and carnosol inhibited the differentiation of preadipocytes into adipocytes ([71857](#)). In vitro research also shows that rosemary extract inhibits digestive enzymes ([98532](#)).

Antioxidant effects

Animal research shows that rosemary extract may have antioxidant effects in aged rats ([71858](#)). Some evidence suggests that the antioxidant effects of phenolic constituents of rosemary may help preserve food ([71725](#), [71745](#)). The antioxidant effects of rosemary extract is primary attributed to the phenolic diterpenes, carnosol and carnosic acid ([18246](#), [71692](#), [71693](#), [71724](#), [71906](#), [71909](#)). Rosmarinic acid also appears to have antioxidant effects ([57869](#)). The antioxidant effects of rosemary

may be weaker than vitamin C but stronger than vitamin E ([71889](#)). Cooking and storage can affect the antioxidant capacity of rosemary ([49470](#)).

Antiparasitic effects

Rosemary leaf extract inhibits the motility of epimastigotes of *Trypanosoma cruzi* in vitro. The trypanocidal effect is attributed to the constituent ursolic acid ([71707](#)).

Antiviral activity

In vitro research shows that rosemary extract, and the constituents carnosol, carnosic acid, and rosmanol, inhibit human immunodeficiency virus (HIV) infection, possibly by inhibiting HIV-1 protease and HIV-1 virus replication ([71663](#), [71912](#)). Rosemary extract has also shown antiviral activity against herpes simplex type 1, herpes simplex type 2, and an acyclovir-resistant strain of herpes simplex type 1 ([68447](#), [68488](#), [71951](#)). Rosemary appears to inhibit herpes simplex virus adsorption not intracellular virus replication ([68447](#)).

Bone effects

Animal research shows that rosemary, rosemary essential oil, and monoterpene components of rosemary inhibit bone resorption when added to the food of rats ([71716](#)).

Cardiovascular effects

Rosemary might also have a positive inotropic effect on the heart and increases coronary blood flow ([2](#)). However, conflicting evidence suggests that rosemary oil might have negative inotropic effects ([91730](#)). In vitro research shows that the rosemary constituent carnosic acid can inhibit tumor necrosis factor (TNF)-alpha-induced migration of vascular smooth muscle cells by inhibiting matrix metalloproteinase (MMP)-9 activity. The activation of MMP-9 and migration of vascular smooth muscle cells are thought to play a role in atherosclerosis development ([71824](#)). Other in vitro research shows that rosemary extract has angiotensin converting enzyme (ACE) inhibitory activity ([67226](#)).

Dermatologic effects

It is not clear how rosemary oil works in alopecia areata, although there is some evidence that it might irritate the skin and increase blood circulation ([2](#), [5177](#)). Rosemary may also increase microcapillary perfusion, increasing hair follicle blood supply ([91729](#)).

Rosemary has also been evaluated for protecting against sun damage. In vitro

research suggests that rosemary extract inhibits UV-induced upregulation of matrix metalloproteinase-1 (MMP-1) gene transcription when applied to skin cells. Upregulation of MMP-1 gene transcription has been shown to be involved in the development of skin photodamage ([71830](#)). Rosemary extract may also protect against UV-induced photodamage by reducing the formation of reactive oxygen species and absorbing and scattering UVB radiation ([91728](#)). The rosemary polyphenols rosmarinic acid, carnosic acid, and carnosol are thought to induce these photoprotective effect ([91728](#)).

Gastrointestinal effects

The essential oil might have a spasmolytic effect on smooth muscle of the gastrointestinal tract and in the ducts of the gallbladder ([4](#)).

Hepatoprotective effects

Animal research shows that rosemary can reduce carbon tetrachloride-induced hepatotoxicity, possibly by reducing oxidative stress ([71672](#), [71697](#), [71703](#), [71870](#), [71893](#)).

Immunomodulatory effects

Rosemary has been shown to increase the production of prostaglandin E2 and reduce the production of leukotriene B4 in human polymorphonuclear leukocytes, as well as inhibit the complement system ([44390](#)).

Neurologic/CNS effects

In humans, rosemary intake for 10 days increases levels of brain-derived neurotrophic factor (BDNF), which is thought to be involved in the regulation of the neuroimmune axis of depression ([105326](#)). Preliminary data suggest that rosemary might inhibit cholinesterases and have a protective effect on dopaminergic neurons ([18246](#), [68163](#), [71839](#)). It may also have antidepressant effects, mediated via the monoaminergic system, and antinociceptive effects, mediated via the endogenous opioid system and inhibited by naloxone ([18246](#), [71856](#)). Cell culture research suggests that rosemary extract may enhance the production of nerve growth factor in glioblastoma cells, which might improve the growth and function of nerve tissue ([71731](#)). Rosemary may also protect neurons from oxidative stress ([71819](#), [71880](#)).

Opioid withdrawal effects

Rosemary may reduce symptoms of withdrawal by inducing GABA. Furthermore, analgesic effects of aqueous and alcoholic extract of rosemary have been

antagonized by naloxone. This may imply its interaction with opioid receptors and its antinociceptive effects ([18246](#), [91727](#)).

Classes

[Antiplatelet Agents](#), [Cytochrome P450 1A1 \(CYP1A1\) Inducers](#), [Cytochrome P450 1A2 \(CYP1A2\) Inducers](#), [Hypoglycemic Agents](#), [Salicylate-Containing Herbs & Supplements](#)

References

[See Monograph References](#)

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